Abhyudaya Pokhrel

Email | GitHub | LinkedIn | Dhulikhel, Kavre, Nepal

Education

Bachelor of Engineering in Computer Engineering
Kathmandu University
Expected 2028

Relevant Coursework

- **Computer Engineering:** Digital Logic, Digital Electronics, Electronics Engineering, Microprocessors & Assembly Language
- **Computer Systems:** Data Structures & Algorithms, Object-Oriented Programming, Computer Programming, Database Management Systems, Communication & Networking
- **Engineering Mathematics:** Calculus & Linear Algebra, Discrete Mathematics, Statistics & Probability, Differential Equations, Numerical Methods
- **Engineering Foundations:** Engineering Projects, Engineering Drawing

Technical Skills

Programming Languages: Python, C/C++, JavaScript, SQL
Machine Learning & Data: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib
Web & Systems: Flask, React, Flutter, SQLAlchemy
Tools: Git, GitHub, Jupyter Notebooks
Soft Skills: Team Collaboration, Technical Communication, Problem Solving

Experience

Machine Translation Intern — *Jul 2026 – Oct 2026*
Information and Language Processing Research Lab

- A three-month internship under the Machine Translation research theme.
- Contributed to a trilingual Nepali–English–Tamang machine translation project.
- Continuing research and co-authoring a research paper on this work (in progress).

Projects

Mikha: Your Legal Eye — *2026*
Flask • Gemma (multimodal) • RAG • edge-tts

- Built a pipeline that transcribes photographed Nepali legal documents and splits them into clauses using the document's own numbering.
- Grounded each clause against the actual text of Nepali law (Labour Act 2017, National Civil Code 2017) via embedding-based retrieval over a cached corpus.
- Generated plain-language, per-clause explanations with risk flags and citations, streamed to the browser as each clause completes; added on-demand text-to-speech and free-form Q&A over the processed document.

Lucid: Multimodal Mental Health Screening System — *2026*
React • FastAPI • Node.js/Express • PostgreSQL • Whisper • MentalBERT • Wav2Vec2

- Built an end-to-end screening platform that transcribes a user's spoken reflection with Whisper and analyzes both text and audio.

- Combined MentalBERT text embeddings with Wav2Vec2 acoustic embeddings via late-fusion to classify mental health condition, likely trigger cause, and depression risk level.
- Implemented the full stack: a FastAPI inference server, a Node/Express + Prisma API and database layer, and a React/Vite frontend.

VanRakshya: Forest Conservation System *2026*
Python • PyTorch • Streamlit • PyDeck

- Developed a multimodal AI system for detecting wildfires, illegal logging, and poaching from drone and CCTV feeds.
- Built an audio classifier capable of detecting chainsaw sounds and gunshots using a dual-model approach.
- Created an interactive threat map with one-click ranger dispatch using Streamlit and PyDeck.

Other Projects

- **ChitraDartaa**: Civic issue reporting app with GPS-tagged images, YOLO-based issue detection, and a Flask dashboard.
- **RiskMapper**: Cascading earthquake-risk simulation for Kathmandu with evacuation-route planning.
- **GameVault**: C++ mini-game arcade (Snake, Pong, Space Invaders) with real-time multiplayer over ENet.
- **XRAI**: Healthcare AI tool for rapid chest X-ray screening in rural clinics; built for KU Hackfest 2025.

Open Source Contributions

- **doc2tex (Co-creator)**: Open-source DOCX ↔ LaTeX converter preserving tables, images, equations, and multilingual content. 56+ GitHub stars and 6 forks.

Leadership & Achievements

- **Vice President, KUCC** (2026 – Present): Leading early initiatives and supporting the execution of the new tenure's activities.
- **Executive Member, KUCC** (2025 – 2026): Helped organize technical workshops, coding events, and peer-learning initiatives for the Kathmandu University computing community.
- **Winner, Build with Gemma: Margadarshan**: Won the Google DeepMind and Kaggle hackathon with Mikha, a legal assistance project.
- **2nd Runner-up, ITMEET 2025 Idea Pitching**: Presented e-Badapatra, a digital public-service concept.
- **Finalist, LOCUS Hack-a-Week 2026**: Developed VanRakshya, a multimodal AI system, under one-week hackathon constraints.
- **Finalist, Infinity Codewave Hackathon**: Advanced to the finals with RiskMapper.
- **KU Hackfest 2025**: Competed with XR-AI, a healthcare AI system for chest X-ray screening.